

Research paper

Energy solutions for data center: Comparative analysis of levelized cost of electricity (LCOE) and recent developments

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ABSTRACT

The rapid growth of artificial intelligence (AI) and cloud computing has dramatically increased global data center energy consumption, challenging existing low-carbon infrastructure development. This study addresses a crucial gap by comparing the Levelized Cost of Electricity (LCOE) for wind, solar, solar plus battery storage, nuclear, and natural gas technologies tailored explicitly for meeting Meta's data center energy demands through 2030. Our custom model incorporates detailed capital and operational costs, capacity factors, site-specific data, and sensitivity analyses of key variables. Results indicate solar plus battery storage as the lowest-cost option (\$25.11/MWh), albeit sensitive to CAPEX and capacity factors. Findings guide policymakers, tech firms, and energy planners toward economically viable and environmentally sustainable energy solutions.

1. Introduction

Rapid advancements in AI, digital services, and cloud computing have significantly driven up global data center energy demands. Achieving a stable, sustainable, and affordable energy supply is now pivotal for technology firms such as Meta, Amazon, and Microsoft. Despite several studies on data center energy optimization, a clear research gap remains in comprehensive comparative techno-economic analyses linked explicitly to policy contexts across diverse global settings. This study explicitly fills this gap by integrating detailed techno-economic modeling and comparative international policy assessments. In particular, the research contributes to the research gap in the following three main avenues:

1. Comprehensive, techno-economic comparison of energy solutions explicitly considering AI-driven growth scenarios.
2. Detailed LCOE analysis tailored to Meta's projected energy demand with robust sensitivity analyses.
3. Integrated policy analysis from diverse global contexts to inform strategic energy procurement.

Internationally, countries, such as Ireland, the US, China, and Brazil, are all taking various policy approaches to data center development and meeting its energy consumption. Spanning across four continents and

including both developing and developed countries, these countries present a diverse mix of policy approaches. While Ireland leverages data centers and its interconnections to the major undersea cables to attract foreign investment, China is using data centers to move capital to its western region with abundant renewable sources for economic development.

The interplay between financial investments by large tech firms, energy generation and grid interconnection, and policy to meet increasing energy demand of data centers provides an important context for future directions. By evaluating cost trends, energy reliability, corporate strategies, and country-specific case studies, this study aims to inform decision-making in both the technology and energy sectors, guiding the transition toward a more resilient and sustainable data center expansion and energy usage.

1.1. Structure of data centers

Data centers are highly specialized facilities designed to house electronic equipment responsible for data processing, storage, and communications. According to the US Environmental Protection Agency (EPA), a data center's core components include servers for processing, storage equipment for data retention, and network equipment for communication. These facilities process, store, and transmit vast amounts of digital information using ICT (Information and

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Communication Technology) infrastructure. To ensure continuous, high-quality performance, data centers employ specialized power conversion and backup systems, alongside environmental controls, to maintain optimal temperature and humidity for their equipment (US Environmental Protection Agency, 2014). Complex infrastructure is essential for maintaining operational reliability, avoiding costly downtime, and protecting valuable data.

2. Literature review

Previous research has examined various aspects of data center energy management, efficiency improvements, and sustainability practices. However, many existing studies lack explicit linkage between techno-economic analyses and international policy implications, highlighting the research gap this study addresses.

The power demand from data centers is expected to increase significantly due to the rise of artificial intelligence (AI) workloads. Goldman Sachs predicts a compound annual growth rate (CAGR) of 15 % for AI-driven power demand between 2023 and 2030. By 2030, data centers could account for 8 % of the total US power demand, compared to around 3 % today. This trend is part of a larger national increase in power demand, which is expected to grow at a 2.4 % CAGR through 2030, up from near-zero growth in the past decade (McKinsey, 2023).

Supporting this growth will require approximately 47 GW of new power generation capacity, with around 60 % of this demand met by natural gas and 40 % from renewable sources like solar and wind. The capital investment needed for this expansion is estimated at \$50 billion (PJM Interconnection, 2023). Potential bottlenecks include lengthy interconnection queues and delays in transmission infrastructure expansion, which could slow the pace of growth (McKinsey, 2023). However, utilities do not currently anticipate major disruptions in power generation. On the flip side, Masanet et al. (2020) argues that while data center energy consumption has increased, thanks to technological and operational efficiency improvements, such increase is relatively small compared to the substantial expansion of data centers.

Hintemann and Hinterholzer (2019) analyzed the past and projected trends in data center energy consumption, addressing key challenges to sustainable operation. Their study highlighted improvements in energy efficiency through IT hardware and software advancements, the adoption of renewable energy sources, and innovative approaches to utilizing waste heat from data centers. Such findings underline the importance of integrating energy-efficient technologies and renewable resources into the design and operation of data centers.

Another prominent area of research focuses on energy savings in green data centers. Zhou, Lou, and Jiang (2019) highlighted the inefficiencies of modern data centers operating at peak load with significant idle or lightly loaded servers and network devices. They developed an energy-saving strategy using the artificial bee colony algorithm for optimizing switch energy consumption in Fat Tree network topologies. Their simulation results indicated significant energy savings and reduced task completion times under varying traffic conditions, emphasizing the effectiveness and scalability of algorithm-driven energy management approaches.

Energy-aware resource management frameworks have also garnered significant attention in addressing energy inefficiencies in IT equipment and facilities. Diouani and Medromi (2019) explored methods to optimize power and performance by scaling virtual machines, adjusting CPU frequencies, and implementing workload consolidation techniques. They presented a comprehensive formula for estimating energy consumption in cloud data centers, providing a valuable tool for quantifying and reducing energy usage through resource allocation strategies.

Parolini et al. (2008) proposed a unified modeling framework for data center energy management that integrates computational performance, thermal dynamics, and power consumption. Their approach uses interconnected component networks to model the spatial distribution of

servers, cooling units, and non-computational components within the data center. The framework employs a Markov Decision Process to coordinate cooling and load management strategies, minimizing the combined cost of power consumption and performance degradation. Their simulation results demonstrated the potential of coordinated control to outperform traditional, independently managed computational and cooling subsystems, suggesting avenues for further research in holistic energy management strategies.

Zhang and Liu (2022) developed a global energy consumption prediction framework for hyperscale data centers. Their framework integrates physical and statistical models to calculate power usage effectiveness (PUE), carbon emissions, and electricity costs. By using hourly meteorological data and energy parameters, they estimated PUE values across 60 regions and analyzed carbon usage effectiveness and electricity costs in China. The framework's predictive accuracy and applicability to regional feasibility studies underscore its potential for guiding energy-efficient HDC development, particularly in the Asia-Pacific region.

Xiang et al. (2023) addressed the dynamic and nonlinear characteristics of energy consumption in large-scale data centers, emphasizing the start-stop relationships and climbing constraints of IT equipment. Their optimization model incorporated both IT and cooling system energy characteristics, leveraging a deep deterministic policy gradient algorithm for effective problem-solving. Simulation results from a 100,000-IT-equipment data center validated the model's practicality and efficiency, offering a scalable solution for minimizing energy consumption in real-world applications.

The environmental impact of data centers is a growing concern, as they consume approximately 1.8 % of U.S. electricity and significant amounts of water for cooling and electricity generation. A study by Shehabi et al. (2021) provided a spatially detailed analysis of the carbon and water footprints of U.S. data centers, revealing that nearly one-fifth of servers operate in water-stressed regions and that power plants supplying these servers often exacerbate regional water stress. These findings highlight the need for targeted strategies to reduce both carbon and water footprints, ensuring sustainable growth of data center infrastructure.

Zong et al. (2024) examined the trends, influencing factors, and classification of data center energy consumption. Their study provided actionable insights into energy-saving measures, particularly for air conditioning systems, and analyzed the energy-saving potential of data centers. The authors emphasized the importance of policy recommendations to promote energy conservation, further reinforcing the role of regulatory frameworks in driving sustainable data center practices.

Collectively, these studies underscore the critical importance of energy optimization in data centers, offering a wide array of methodologies ranging from algorithmic approaches and resource management frameworks to predictive models and policy interventions. The integration of these strategies is essential to address the increasing energy demands of data centers while minimizing their environmental and economic impact.

2.1. Peak energy consumption

Peak energy consumption in data centers typically occurs during business hours—between 10 AM and 6 PM—when cloud services and data processing demands are highest. AI workloads, which are computationally intensive, add further strain during these hours. Data centers are significant contributors to grid demand during these peak periods, particularly in summer months when cooling requirements are most substantial (PJM Interconnection, 2023).

To manage peak energy usage, many data centers have implemented peak-shaving programs and energy efficiency measures. These strategies are crucial in mitigating the impact of high demand periods on the grid and ensuring more sustainable operations (McKinsey, 2023).

Cooling uses the most amount of energy in data centers, attributing

Table 1
Data center cooling technologies.

	Technology	System efficiency & methods
Air cooling	Common due to its simplicity and low cost, but less energy efficient. Various levels include room-level (raised floors), row-level (near server rows), and rack-level (cooling within racks). Challenges include inefficiencies like hot air recirculation and cold air bypass, which can be mitigated using containment systems.	Low maintenance but inefficient in high-heat environments. Includes different levels of cooling (room, row, and rack), with challenges related to airflow and space constraints.
Liquid cooling	More efficient, especially in high-heat environments. It includes indirect cooling (such as single-phase, two-phase, and heat-pipe methods) and direct cooling (such as pool-boiling and spray-boiling). Liquid cooling is more effective for high-density data centers but comes with maintenance and leakage risks.	More efficient, reducing power usage, and especially suited for high-density setups. Indirect cooling is less risky, while direct methods offer high efficiency but can cause damage if components are immersed in coolant.
Free cooling	Utilizes natural cold sources to reduce energy consumption, particularly in areas with cooler climates. It can be direct (using cold air) or indirect (using heat exchangers to transfer heat without direct exposure to outside conditions). Challenges include the risk of contaminants in direct cooling and the maintenance costs of heat exchangers in indirect systems.	Most energy-efficient, leveraging the environment’s natural cold. Direct free cooling is cheaper but riskier, while indirect cooling offers better protection from environmental factors but at higher maintenance costs.

Table 2
LCOE and Assumptions.

Assumptions	Wind	Nuclear	Solar	Natural gas	Solar + Battery Storage
Capital Cost per kW (\$)	2100	16710	1400	850	1800
Fixed O&M Cost per kW per year (\$/kW-yr)	40	140	20	20	30
Variable O&M (t\$/MWh)	0	5	0	5	8
Heat rate (MMBtu/MWh)	0	10	0	7	0
Fuel price (\$/MMBtu)	0	0.85	0	3	0
Capacity Factor (%)	30%	85 %	25 %	65 %	48 %
Decommissioning Cost (% of CapEx)	3%	10.00 %	2 %	1 %	5 %
Lifetime (Years)	25	60	28	30	25
LCOE (\$/MWh)	48.14	73.83	32.42	37.69	25.11

Table 3
Sensitivity analysis.

Scenario	Technology	LCOE (\$/MWh)	% difference from base case	Result
Base case	Natural gas	36.97	-	
	Solar+Battery storage	25.11	-	
CAPEX + 10 %	Natural gas	37.40	1 %	Solar+Battery Storage is more sensitive
	Solar+Battery storage	26.91	7 %	
CAPEX − 10 %	Natural gas	36.54	− 1 %	Solar+Battery Storage is more sensitive
	Solar+Battery storage	23.32	− 7 %	
Capacity factor + 5 %	Natural gas	36.41	− 2 %	Solar+Battery Storage is more sensitive
	Solar+Battery storage	22.74	− 9 %	
Capacity factor − 5 %	Natural gas	37.62	2 %	Solar+Battery Storage is more sensitive
	Solar+Battery storage	28.03	12 %	

50 % of the total energy consumption, followed by server & storage and power conversion, at 26 % and 10 %, respectively (Wu, Jin, & Yao, 2017).

According to Zhang et al. (2021), Data center cooling can be categorized into two main subsystems: the Mechanical Refrigeration Sub-System (MRSS) and the Terminal Cooling Sub-System (TCSS). The MRSS uses equipment like chillers, pumps, and cooling towers to cool the data center, while the TCSS absorbs heat inside the data center and transfers it outside. In detail, cooling technologies include:Table 1

3. Method

3.1. Levelized cost of electricity of power generation sources to meet energy demand of meta’s data centers

This study estimates the Levelized Cost of Electricity (LCOE) for five generation technologies—wind, solar, solar combined with battery

storage, nuclear, and natural gas—to assess their viability in meeting Meta’s projected data center energy demand through 2030. These generation methods, A quantitative model was developed in Microsoft Excel to simulate LCOE outcomes based on techno-economic assumptions.

Meta’s data center energy usage is used as a case study because the company operates numerous large-scale data centers globally, consistently publishes detailed sustainability and energy consumption reports, and actively engages in renewable energy procurement. This makes Meta’s energy strategy highly relevant for broader insights applicable to major global technology companies. Additionally, Meta provides publicly accessible, detailed historical and projected energy usage data, enabling robust analysis and validation of assumptions.

The generation technologies selected for this study (wind, solar, solar plus battery storage, nuclear, and natural gas) were chosen to represent a balanced mix of renewable, low-carbon, and conventional power sources that are most relevant to the current and emerging energy strategies of large-scale data center operators. Wind and solar are mature, widely deployed renewable technologies that align with sustainability goals but face challenges due to their intermittent nature. To address this, solar combined with battery storage was included to evaluate the economic and operational implications of integrating storage into renewable systems. Nuclear energy, particularly with renewed interest in small modular reactors, offers a reliable, carbon-free baseload option despite its high capital costs. Natural gas, while carbon-intensive, remains a key transitional fuel due to its dispatchability and cost-effectiveness. Collectively, these technologies reflect the range of options actively considered by companies like Meta and allow for a comprehensive comparison of trade-offs in cost, reliability, and environmental impact.

The LCOE formula used follows the conventional structure applied in energy economics (EIA, 2023), where LCOE is calculated as the ratio of the net present value of total life-cycle costs to the total electricity output over the plant’s lifetime. This includes capital expenditures (CapEx), fixed and variable operations and maintenance (O&M) costs, fuel costs (for dispatchable sources), decommissioning expenses, and the technology-specific capacity factor and lifespan. While tax credits and

policy instruments played important role in the LCOE calculations, with a potential for them to no longer exist, this LCOE calculation assesses costs without tax credits. Where site-specific data were available (e.g., NREL maps for solar irradiance, regional wind speed maps), adjustments were made to national averages to better reflect local realities and improve model accuracy.

The mathematical formulation of the LCOE calculation is as follows:

$$LCOE = \sum_{t=1}^n \frac{(I_t + O_t + F_t + D_t)}{(1+r)^t} \div \sum_{t=1}^n \frac{E_t}{(1+r)^t}$$

Where :

I_t : Capital investment expenditure in year t

O_t : Fixed and variable operation and maintenance costs in year t

F_t : Fuel costs in year t

D_t : Decommissioning cost in year t

E_t : Electricity generated in year t

r : Discount rate (set to 0 for undiscounted analysis)

n : Economic lifetime of the project (years)

Data inputs were drawn from public and industry sources, including capital and operating cost assumptions from Lazard's Levelized Cost of Energy Analysis – Version 17.0 (Lazard, 2024), Meta's historical and projected data center energy consumption (Meta Sustainability Report, 2024), and electricity demand growth projections from McKinsey & Company (McKinsey, 2024).

The total electricity demand from Meta's data centers was projected to reach approximately 61,735,467 MWh by 2030, based on extrapolations from McKinsey and Meta reports. Adjusted installed capacity requirements for each generation type were estimated by dividing the total energy demand by the annual energy generation of each technology, accounting for respective capacity factors and a 15 % reserve margin.

A sensitivity analysis was conducted to evaluate the impact of ± 10 % variations in capital costs and ± 5 % variations in capacity factors on LCOE for two selected technologies: natural gas and solar + battery storage. These ranges were chosen based on realistic uncertainty bounds observed in past technology cost evolutions (NREL ATB, 2024). All sensitivity calculations followed the same LCOE formulation, with each scenario compared to the base case to quantify relative sensitivity.

This method builds upon the general LCOE calculation framework widely used in power sector planning and policy studies (EIA, 2023; Lazard, 2024), with modifications tailored to Meta's projected data center energy demand. Exclusions (e.g., tax credits and discount rates) were deliberately chosen for baseline simplicity, allowing direct comparative assessments of inherent technology costs without distortions from variable financial incentives or tax structures, given the current policy uncertainty around tax credits and financing structures in the US.

4. Results

Based on the assumptions and inputs in the model, the calculated Levelized Cost of Electricity (LCOE) for each generation source to meet Meta's data center energy demand through 2030 is as follows:

The analysis of LCOE for different generation technologies highlights the trade-offs between capital costs, operational expenses, and efficiency. Nuclear power has the highest capital cost (\$16,710/kW) and fixed O&M costs (\$140/kW-yr), driving its LCOE to \$73.83/MWh despite its high capacity factor (85 %) and long lifespan (60 years). Wind

and solar, with lower capital costs (\$2100/kW and \$1400/kW, respectively), exhibit lower LCOEs of \$48.14/MWh and \$32.42/MWh, though their lower capacity factors (30 % and 25 %) limit generation potential. Natural gas, with a moderate LCOE of \$37.69/MWh, benefits from a high capacity factor (65 %) and low capital costs (\$850/kW), though fuel costs (\$3/MMBtu) contribute significantly to total expenses. The most cost-effective option is solar with battery storage, which balances capital costs, avoids fuel expenses, and achieves an LCOE of \$25.11/MWh, making it a competitive choice for future energy generation.

Other analysis, such as Lazard's, (2024) LCOE calculations, present findings that are relatively higher than the LCOE analysis based on varying input assumptions, such as variable costs, capacity factors, and discount rates. While the Lazard LCOE calculation incorporates discounted cash flow project finance methods as well as tax credits, the model presented in this article doesn't necessarily consider those for simplicity. However, there are similarities between the two analysis and can be viewed comparatively.

4.1. Sensitivity analysis

The sensitivity analysis evaluates the impact of capital expenditure (CAPEX) and capacity factor variations on the Levelized Cost of Energy (LCOE) for natural gas and solar with battery storage. A 10 % increase in CAPEX raises LCOE by 1 % for natural gas and 7 % for solar with battery storage, while a 10 % decrease lowers it by 1 % and 7 %, respectively, indicating that solar with battery storage is more sensitive to CAPEX fluctuations. Similarly, a 5 % increase in capacity factor reduces LCOE by 2 % for natural gas and 9 % for solar with battery storage, whereas a 5 % decrease raises LCOE by 2 % and 12 %, respectively, further demonstrating that solar with battery storage is more sensitive to operational performance variations. These results highlight the greater financial vulnerability of solar with battery storage to changes in both CAPEX and capacity factor compared to natural gas.

5. Discussion

5.1. Realistically meeting energy demand of data centers

The analysis demonstrates that solar with battery storage might be the cheapest generation option for meeting Meta's data center energy demand with an LCOE of \$25.11/MWh. However, when considering reliability, natural gas might be a more viable option given the relative novelty of battery storage. However, reliance on natural gas raises concerns about carbon emissions and regulatory risks, making renewable and low-carbon alternatives increasingly relevant. Nuclear, despite its high capacity factor (85 %), remains the most expensive option at \$73.83/MWh, primarily due to substantial capital expenditures. These findings underscore the trade-offs between cost, reliability, and sustainability in energy procurement decisions for Meta's data centers.

5.2. Capacity expansion: technology and energy companies

The increasing energy consumption of data centers is prompting large technology companies to directly engage with energy companies and utilities to secure reliable, low-carbon energy sources. This strategic shift reflects the growing need for tech firms to manage energy risks while ensuring that their operations align with sustainability goals. A key aspect of this transformation is the exploration of nuclear energy, particularly small modular reactors (SMRs), by tech giants like Amazon and Microsoft. Their interest in nuclear power underscores the tech sector's commitment to carbon-free energy and illustrates a potential pathway for nuclear energy adoption across industries.

Simultaneously, these companies are leveraging their resources and influence to foster collaboration with utilities, resulting in the development of co-located energy parks that integrate renewable energy with data center operations. This model not only meets immediate energy

demands but also contributes to broader sustainable infrastructure efforts. However, the accelerating demand for energy from data centers presents challenges for utilities, which must develop new rate designs, collaborative models, and innovative solutions to balance grid stability and cost distribution, exemplified by the recent agreement between Indiana Michigan Power (I&M) and major tech companies. Despite these efforts, the rapid pace of demand growth is putting pressure on grid infrastructure, further complicating the energy transition.

Amazon, for instance, is investigating the potential of small modular reactors to support its operations, with the aim of achieving net-zero carbon emissions (Amazon, 2024). This approach represents a significant shift in how tech companies view their energy sourcing strategies, potentially paving the way for more widespread adoption of nuclear power in the tech sector. Similarly, Microsoft is considering an agreement with Constellation to purchase power from the Three Mile Island nuclear plant, further underscoring the tech industry's interest in reliable, carbon-free energy sources (Reuters, 2024, 2024).

Concurrently, major technology firms are actively collaborating with utilities to develop innovative solutions for power distribution and integration. Companies such as Google, Microsoft, and Amazon are entering into interconnection agreements with utilities, facilitating the development of co-located energy parks that integrate renewable energy sources directly with data center operations (Utility Dive 2024a; Utility Dive 2024b). These partnerships not only address the immediate energy needs of data centers but also contribute to the broader transition towards sustainable energy infrastructure. By leveraging their significant resources and influence, tech companies are playing a pivotal role in shaping the future of energy production and distribution, potentially accelerating the adoption of renewable and low-carbon energy sources across various sectors.

In the energy systems scheme, utilities are developing new rate designs and exploring collaborative models with large customers to ensure equitable cost distribution and sustainable infrastructure development. For instance, some utilities are implementing long-term contracts with minimum charges, while others are facilitating direct customer contributions to clean energy projects and grid improvements (Reuters, 2024, 2024). For example, Duke Energy, along with Amazon, Google, Microsoft, and Nucor, announced agreements on May 29, 2024, to develop *Accelerating Clean Energy (ACE) tariffs*, which would allow large energy consumers to invest in emerging clean technologies like advanced nuclear and long-duration storage, reducing project risks and costs (Duke Energy, 2024). A key feature, the *Clean Transition Tariff (CTT)*, would enable customized carbon-free energy portfolios for commercial and industrial customers, supporting grid decarbonization while requiring regulatory approval in North and South Carolina (Duke Energy, 2024).

In addition, a recent agreement between Indiana Michigan Power (I&M), an American Electric Power (AEP) utility, and major tech companies including Amazon, Google, and Microsoft, exemplifies the evolving strategies to address this surge in energy demand (Utility Dive 2024a). This settlement, filed with the Indiana Utility Regulatory Commission, outlines new interconnection rules for large load customers and represents a proactive response to the projected doubling of I&M's peak demand from approximately 4000 MW in 2023–8000 MW by 2030. The agreement encompasses several key provisions designed to balance the needs of new large customers with the interests of existing ratepayers and grid stability. It mandates long-term financial commitments from new large load customers, establishes minimum 12-year initial contracts, and provides structured capacity reduction options. Furthermore, it commits the utility to studying grid-enhancing technologies and engaging in stakeholder discussions about interconnection processes (Utility Dive 2024a). This collaborative approach not only addresses the immediate energy needs of data centers but also contributes to broader grid modernization efforts.

However, the rapid increase in demand has outpaced many utilities' preparedness, with some potentially needing to increase their annual generation by more than 25 % over a three-year period. This

acceleration is exacerbating existing challenges, particularly in grid infrastructure development, which is hampered by lengthy planning and permitting processes.

5.3. Data centers location siting decision and undersea cables

Data centers are strategically located to be connected to undersea cables transmitting information. Spanning over 1400,000 km, approximately 485 undersea cables carry data at speeds of multiple terabits per second, making them the fastest and most efficient data transfer method in existence. Undersea cable operations have traditionally been dominated by telecommunication companies. As of 2017, the US telecom giant AT&T alone operated 90 undersea cable systems stretching over 700'000 km. However, in recent years, tech giants such as Google, Microsoft, and Meta have emerged as dominant players in this space. Notable projects include:

1. MAREA, a 6605 km transatlantic cable developed through a partnership between Meta and Microsoft, which connects the US with Spain. As of 2019, it had the highest data transmission capacity in the world at 200 terabits per second.
2. Another significant project is ANJANA, a 7121 km transatlantic fiber-optic cable owned and operated by Meta and its subsidiaries, including Edge Network Services Limited. ANJANA connects Myrtle Beach, South Carolina, US with Santander, Spain.
3. Additionally, the JUNO cable spans 11,700 km with three segments connecting the US and Japan. It is owned by Seren Juno Network Co., Ltd., a joint venture established in 2022 involving NTT Ltd. Japan Corporation, Mitsui & Co., Ltd., PC Landing Corp., and JA Mitsui Leasing, Ltd.

6. Conclusion

This study explores the intricate dynamics between data center energy demand, cost considerations, technological advancements, and global policy strategies. The LCOE analysis reveals that while natural gas offers a relatively cost-effective solution for powering data centers like Meta's, the long-term environmental implications necessitate a transition towards renewable and low-carbon alternatives. Solar with battery storage, under provided assumptions, presents the cheapest option along with wind and solar being viable, yet their intermittency requires innovative grid management and more advanced storage solutions to ensure consistent reliability. Nuclear power, despite its high upfront costs, offers a stable and carbon-free energy source that warrants further consideration, particularly with advancements in small modular reactor technology.

The increasing engagement of tech companies in direct energy procurement signifies a paradigm shift, fostering collaboration with utilities and driving the development of co-located renewable energy parks. This proactive approach addresses the immediate energy needs of data centers and contributes to the broader transition towards sustainable energy infrastructure. However, the rapid pace of demand growth poses significant challenges for utilities, requiring innovative rate designs, collaborative models, and accelerated grid modernization efforts.

Globally, diverse strategies are being adopted to foster data center growth, ranging from Ireland's investment-friendly policies to China's strategic relocation initiatives focusing on renewable energy sources and economic development and Brazil's ambition to build a data center city. These case studies highlight the importance of tailoring approaches to local energy landscapes, infrastructure development, and regulatory frameworks.

As these current global dynamics shape data center and energy demand growth, these challenges lay ahead:

Grid modernization: Upgrading and expanding grid infrastructure to accommodate the increasing energy demands of data centers while maintaining stability and reliability remains a significant hurdle.

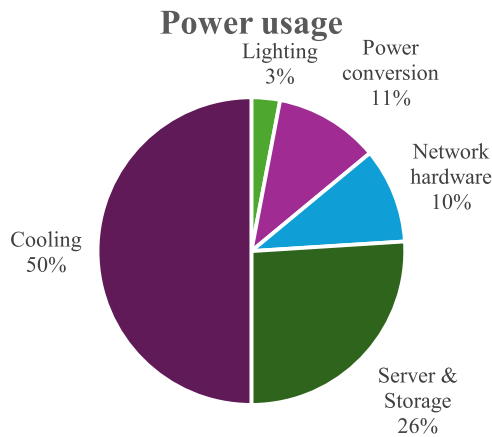


Fig. 1. Data center power usage.

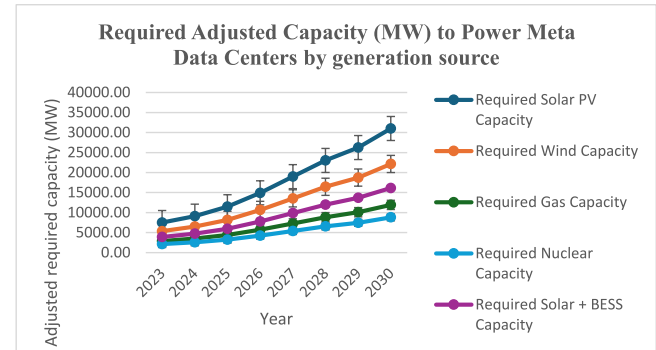


Fig. 2. Required adjusted capacity (MW).

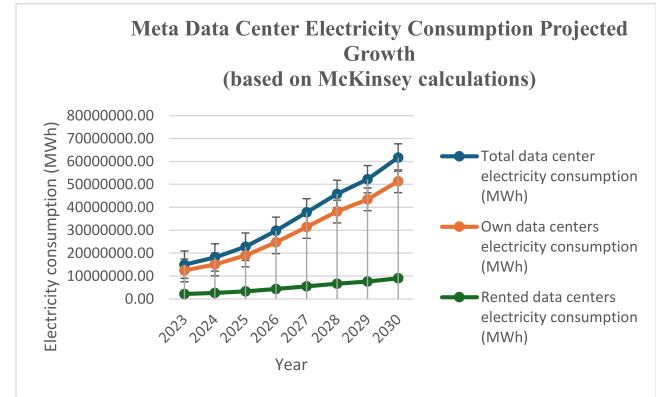


Fig. 3. Meta data center consumption projected.

Streamlining planning and permitting processes for grid enhancements is crucial.

Intermittency of renewables: Addressing the intermittency challenges associated with wind and solar power requires investments in energy storage solutions, such as advanced battery technologies and pumped hydro storage, as well as improved forecasting and grid management capabilities.

Cost competitiveness of nuclear: Reducing the capital costs of nuclear power through technological innovation, standardization, and policy incentives is essential to enhance its competitiveness and facilitate wider adoption.

Sustainable siting: Strategically locating data centers to leverage access to undersea cables and renewable energy resources while



Fig. 4. Submarine Cable (Getty Images).

minimizing environmental impact and community disruption will be crucial for sustainable growth.

Policy and regulatory frameworks: Developing clear and consistent policy and regulatory frameworks that support sustainable data center development, incentivize renewable energy adoption, and ensure equitable cost distribution is vital for fostering a resilient and sustainable digital infrastructure.

Workforce development: Meeting all of these challenges requires a workforce that is trained in the design, development, management, and maintenance of data centers and associated energy infrastructure.

Addressing these challenges will require concerted efforts from technology companies, utilities, policymakers, and researchers to ensure that data centers can continue to support the advancement of AI and digital expansion in a sustainable and responsible manner. Fig. 1

Author statement

Tsedensodnom Uranbold conceptualized the study, developed the model, and performed the techno-economic analysis. Luana Marangon Lima supervised the research and contributed to the review and editing of the manuscript. Both authors contributed to the policy analysis and approved the final version of the manuscript.

CRediT authorship contribution statement

Luana Marangon Lima: Writing – review & editing, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Tsedensodnom Uranbold:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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